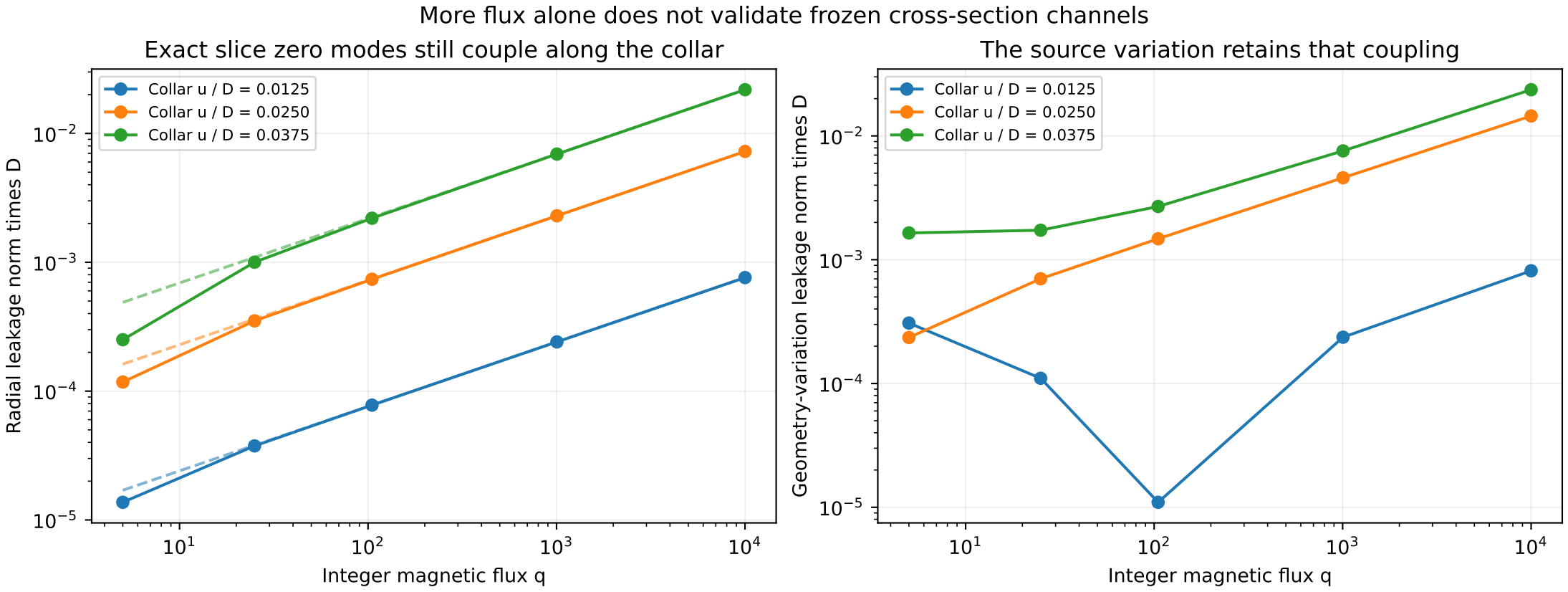
More flux alone does not validate frozen cross-section channels
Exact slice zero modes still couple along the collar
The source variation retains that coupling
Collar u / D = 0.0125
Collar u / D = 0.0250
Collar u / D = 0.0375
Radial leakage norm times D
Geometry-variation leakage norm times D
Integer magnetic flux q
Integer magnetic flux q
Dashed curves: local square-root-q limits. One mode per q; no global source bound or atom transit.